Business Case Template

1. Executive Summary

This business case proposes the development of a web-based, mobile-responsive intelligent decision support system for early disease risk detection in companion animals. The system leverages Machine Learning for predictive risk classification and an Agentic AI reasoning layer to generate contextual, explainable preventive recommendations.

Pet owners input symptom and behavioral observations through structured forms or an AI-assisted chatbot interface. The system processes this data through a cloud-based ML inference service to classify disease risk into Low, Medium, or High categories. Based on risk level, the Agentic AI layer autonomously generates actionable guidance including home care advice, veterinary consultation recommendations (24–48 hour thresholds), vaccination alerts, and emergency escalation notifications.

The platform integrates geolocation-based veterinary clinic discovery using Google Maps API and cross-references predicted disease types with clinic specialization and appointment availability stored within Firebase Firestore.

The proposed solution enhances early intervention, reduces treatment costs, supports preventive veterinary care, and establishes a scalable AI-driven digital health platform for the Sri Lankan pet healthcare ecosystem.

2. Introduction and Summary

a. Rationale for Preferred Option

The preferred option is to develop a web-based mobile, AI decision support system using Machine Learning and Agentic AI, rather than a purely informational app or a clinician-only tool. This option is selected because it:

- Enables early disease risk prediction using symptom data.
- Provides autonomous reasoning and guidance
- Preventive care integration
- Integrates vaccination tracking, emergency alerts, and real-time veterinary discovery.

- Is accessible to pet owners, not limited to veterinarians.
- Aligns directly with modern healthcare trends in AI-driven preventive care.

Alternative options such as static information systems or vet-only tools were rejected due to their lack of autonomy, real-time risk prediction, and owner engagement.

b. Current Business Process

Pet owners typically rely on:

- Internet searches or informal advice
- Delayed veterinary visits after symptoms worsen
- Manual tracking of vaccinations and health history

Veterinary healthcare is largely reactive rather than preventive. Existing systems:

- Focus on appointments, reminders, or clinical diagnostics,
- Do not offer early risk prediction or autonomous guidance at the owner level.

There is no widely adopted mobile system that provides AI-driven, personalized, preventive decision support for pet owners.

c. Description of the Problem

Companion animal health management currently relies heavily on pet owners' ability to recognize early symptoms and seek timely veterinary care. However, early-stage diseases in dogs and cats often present with non-specific behavioral or physiological changes such as reduced appetite, lethargy, or altered urination patterns, which are difficult for non-experts to interpret. Due to limited awareness and lack of medical knowledge, pet owners frequently overlook these early warning signs, resulting in delayed veterinary consultation.

Existing digital pet care applications primarily focus on vaccination reminders, health record storage, or appointment booking, and do not provide intelligent early risk assessment or preventive decision support. The absence of an accessible, data-driven system for early disease risk identification leads to late diagnosis, increased treatment costs, reduced recovery outcomes, and compromised animal welfare. This highlights a critical gap for a proactive, user-friendly solution that supports early detection and timely intervention at the pet owner level.

d. Opportunity

The increasing number of companion animal owners and growing awareness of preventive pet healthcare present a significant opportunity for a digital early disease risk assessment solution. Currently, pet owners rely primarily on reactive veterinary consultations, often after symptoms become severe, which leads to higher treatment costs and poorer health outcomes.

This project creates an opportunity to develop a proactive, AI-driven decision support platform that enables early risk awareness based on observable symptoms and behavioral changes. By providing accessible, text-based risk assessment and guided recommendations, the system can bridge the gap between everyday pet observations and timely veterinary intervention.

From a business perspective, the solution offers the following opportunities:

- Growing market demand for digital pet health and telehealth solutions
- Potential integration with veterinary clinics, hospitals, and pet care providers
- Opportunity to develop a scalable mobile health platform for the Sri Lankan and regional markets
- Support for preventive healthcare services, reducing long-term treatment costs
- Potential for future expansion into subscription-based services, teleconsultation, and health monitoring ecosystems

This positions the project as a value-creating digital health innovation that improves animal welfare while addressing an emerging market need in the pet healthcare sector.

e. Objectives

Primary Objective

- To design and implement a scalable, AI-driven, web-based decision support architecture that integrates machine learning prediction and agentic reasoning for early disease risk identification and preventive veterinary guidance.

Specific Objectives

- To provide an easy-to-use platform for pet owners to record symptoms and behavioral changes.

- To generate early disease risk predictions using machine learning based on user-provided data.
- To deliver actionable preventive recommendations through an Agentic AI decision layer.
- To support timely veterinary consultation by providing location-based clinic and surgeon discovery.
- To improve preventive healthcare through vaccination tracking and automated reminders.
- To maintain a centralized digital health history for continuous monitoring of pet health.
- To increase early awareness and reduce delayed treatment through proactive alerts and notifications.
- To ensure accessibility and usability for non-technical users through a simple, mobile-responsive interface.

Business Value Alignment

- Reduces treatment costs through early intervention.
- Improves animal health outcomes and owner satisfaction.
- Creates a scalable digital health support solution for the growing pet care market.
- Supports data-driven preventive veterinary services.

f. Project Scope

In Scope

System Users

- Dog and cat owners (primary users)
- Veterinary clinics (information listing and availability)

Core Functional Scope

- User registration, authentication, and profile management

- Pet profile creation (species, age, breed, medical history)
- Symptom and behavioral data entry through structured text-based forms
- Machine learning–based early disease risk prediction
- Agentic AI–based interpretation and personalized preventive guidance
- Vaccination schedule tracking and automated reminders
- Emergency risk alerts with recommended actions
- Veterinary clinic and surgeon discovery with location and available time slots
- Health history storage and progress tracking
- Notification and feedback mechanism

Technology Scope

- Web-based, mobile-responsive application built using React.js with Tailwind CSS for modern UI styling and responsive design
- Backend API layer using Node.js with Express.js for request orchestration
- Cloud-based database using Firebase Firestore for secure storage of pet health records and clinic metadata
- Machine Learning inference service implemented in Python FastAPI integrated with Google Vertex AI
- Agentic AI reasoning layer using LangChain, LangGraph, and LangSmith
- Integration with Google Maps API for geolocation-based clinic discovery
- Firebase Cloud Messaging for real-time alerts and vaccination reminders

g. System Operational Flow

The system follows a multi-layered architecture integrating user interaction, machine learning inference, agentic reasoning, and external service orchestration.

Step 1 – Symptom Entry

The pet owner inputs symptom and behavioral observations via:

- Structured digital forms
- AI-assisted conversational chatbot

Chatbot inputs are converted into structured symptom attributes using natural language processing before being forwarded to the backend.

Step 2 – Data Storage

All submitted symptom data is first stored in Firebase Firestore under the pet's longitudinal health record.

Step 3 – ML Risk Prediction

The backend sends structured symptom data to the Python FastAPI ML service hosted via Google Vertex AI.

The trained model predicts:

- Probable disease category
- Risk confidence score
- Risk level classification:
    - High Risk – Vet visit within 24 hours
    - Medium Risk – Vet visit within 48 hours or if symptoms persist
    - Low Risk – Home monitoring recommended

Step 4 – Agentic AI Reasoning Layer

The ML output is forwarded to the Agentic AI module (LangGraph-based workflow), which:

- Interprets prediction confidence
- Checks vaccination history
- Evaluates emergency indicators
- Generates contextual recommendations

Step 5 – Veterinary Recommendation Mapping

If veterinary consultation is required:

- User geolocation is obtained (with permission)
- Google Maps API retrieves nearby clinics
- Firestore clinic collection is queried for:
    - Specializations

    - Surgeons
    - Available time slots
- Agent ranks clinics based on disease type and urgency

Step 6 – Vaccination Logic Integration

If predicted disease is vaccine-preventable:

- Vaccination history is checked
- Overdue vaccines are identified
- Preventive vaccination reminders are generated

Step 7 – Notification System

Firebase Cloud Messaging sends:

- Emergency alerts
- Vaccination reminders
- Appointment confirmations

This architecture ensures seamless integration between predictive modeling, autonomous reasoning, and real-world veterinary service discovery.

h. Business Benefits

Operational Benefits

- Enables early identification of potential health risks, reducing emergency veterinary visits.
- Supports preventive care through vaccination tracking, reminders, and health monitoring.
- Improves decision-making for pet owners through AI-driven risk assessment and guidance.
- Provides centralized digital health records for pets, improving continuity of care.

Financial Benefits

- Reduces treatment costs by detecting diseases at early stages.

- Minimizes unnecessary veterinary visits through preliminary risk evaluation.
- Creates potential revenue opportunities through partnerships with veterinary clinics, pharmacies, and pet service providers.
- Scalable cloud-based solution with low operational cost compared to traditional healthcare systems.

Customer/User Benefits

- Improves awareness and confidence of pet owners in managing pet health.
- Provides 24/7 accessible health support through a mobile web platform.
- Saves time by offering nearby veterinary discovery and appointment availability.
- Enhances user engagement through personalized alerts, reminders, and recommendations.

i. Project Costs

Cost Category	**Description**	**Estimated Cost (LKR)**
Application Development	Frontend, backend, API integration	
Machine Learning & AI	Model training, experimentation	
Cloud Infrastructure	Firebase, Vertex AI, hosting	
APIs & Tools	Google Maps API, notifications	
Data Collection	Survey design, data labeling	
Testing & Evaluation	Usability testing, validation	
Contingency	Unforeseen costs	

j. Assumption

The following assumptions are made for planning and evaluation of the proposed system:

- Pet owners are willing to provide accurate symptom and behavioral data through surveys and system usage.

- Sufficient primary and secondary data will be available to train and validate the machine learning models.
- Users have access to smartphones and internet connectivity to use the web-based mobile application.
- Veterinary clinics will provide publicly available location and contact information.
- The selected technology stack (React, Node.js, Firebase, AI services) will support system scalability and performance.
- Users will follow system guidance as a support tool, not as a replacement for professional veterinary diagnosis.

k. Analysis of Potential Impact on Staff and Business

Impact on Business

- Enables entry into the digital pet healthcare and preventive care market.
- Supports future commercialization through subscriptions, premium features, and partnerships.
- Strengthens brand positioning as an innovative AI-driven healthcare solution.
- Improves data-driven decision-making using collected health and usage data.
- Provides scalability through cloud-based architecture.

Impact on Staff / Operations

- Reduces manual effort through automation of risk assessment, guidance, and notifications.
- Supports veterinarians by enabling early case identification.
- Requires technical staff for system maintenance, model monitoring, and updates.
- Provides technical skill development in AI, cloud, and Agile practices.
- Requires minimal operational staff due to high system automation.

l. Analysis of Potential Impact on Technology

- Introduces AI-driven, cloud-based web application architecture.
- Requires integration of React.js, Node.js, Firebase, Python ML services.
- Requires ongoing model training, monitoring, and performance optimization.
- Increases dependency on third-party services (Google Maps API, Firebase, Vertex AI).
- Requires strong security controls for authentication, data privacy, and system reliability.
- Agile and cloud deployment enable scalability, continuous updates, and future enhancements.
- Requires specialized technical expertise in AI, cloud computing, and full-stack development.

m. Other Issues

- Ethical deployment of AI in health advisory systems requires transparency and explainability of recommendations.
- The system must clearly communicate that outputs are probabilistic risk assessments, not medical diagnoses.
- Continuous model monitoring and retraining are required to mitigate concept drift.
- Data governance policies must ensure secure handling of pet health and location data.
- Clinic time-slot data may initially rely on structured manual or periodic updates rather than direct hospital system integration.

n. Implementation Plan

The implementation of the Agentic AI-driven pet health decision support system will follow an Agile, phased deployment strategy. The project will progress through iterative development cycles, ensuring continuous validation and risk mitigation prior to full-scale rollout.

Phase	**Key Activities**	**Deliverables**	**Duration**
Phase 1: Planning & Requirement Finalization	Requirement analysis, technical architecture design, dataset preparation	Approved BRD, System Architecture	2 weeks
Phase 2: Data Preparation & ML Development	Data cleaning, feature engineering, model training & validation	Trained ML model with evaluation metrics	4 weeks

Phase 3: Agentic AI Layer Development	ACL integration, Model Context Protocol setup, reasoning workflow design	Functional Agentic Decision Module	3 weeks
Phase 4: Application Development	Frontend (React), Backend (Node.js), API integration	Working Web-based Mobile Application	4 weeks
Phase 5: Testing & Quality Assurance	Functional testing, usability testing, model performance testing	Tested & validated system	2 weeks
Phase 6: Pilot Deployment	Limited user rollout, feedback collection	Pilot performance report	2 weeks
Phase 7: Final Deployment & Monitoring	Production deployment, monitoring, maintenance setup	Fully operational system	Ongoing

3. Approach

a. Financial Metrics

The financial evaluation of the proposed Agentic AI-based pet health decision support system will be assessed using the following metrics:

Financial Metric	**Purpose**	**Application to This Project**
Cost–Benefit Analysis (CBA)	Compares total project costs against expected benefits	Evaluates development and operational costs against expected revenue (subscriptions, partnerships) and social value
Return on Investment (ROI)	Measures financial return relative to investment	Determines profitability potential over 3–5 years
Net Present Value (NPV)	Assesses long-term financial viability	Evaluates future projected cash flows from subscriptions and partnerships

Break-even Analysis	Identifies when revenues equal costs	Estimates user base required to recover development and operational costs
Total Cost of Ownership (TCO)	Calculates full lifecycle cost	Includes development, hosting, maintenance, AI services, and updates

b. Assessment of Impact on Privacy

The system processes user-provided pet health data and location data. Privacy assessment includes:

Area	**Consideration**	**Mitigation Strategy**
Personal Data	User login credentials and contact details	Secure authentication and encrypted storage
Pet Health Data	Symptom and behavioral records	Stored securely; used only for risk prediction
Location Data	Veterinary clinic discovery	Accessed only with user permission
Data Storage	Cloud-based database (Firebase)	Role-based access control and secure API endpoints
Regulatory Compliance	Data protection standards	Adherence to applicable data protection principles

c. Criteria for Evaluating Alternatives

The proposed solution was evaluated against possible alternatives (e.g., reminder-only apps, chatbot-only systems, clinical AI systems) using the following criteria:

Evaluation Criterion	**Description**

Functional Coverage	Ability to support early disease risk prediction and preventive guidance
Technical Feasibility	Compatibility with available AI and web technologies
Cost Efficiency	Development and operational cost requirements
Scalability	Ability to support increasing users and data volume
User Accessibility	Ease of use for non-technical pet owners
Innovation Level	Degree of autonomous reasoning and explainability
Integration Capability	Ability to integrate ML, Agentic AI, and clinic discovery

The proposed Agentic AI-driven system was selected as the preferred option due to its comprehensive functionality, scalability, preventive focus, and intelligent decision-support capabilities.

4. Key Selection Criteria

a. Weighting

The following criteria are used to evaluate and select the most appropriate solution alternative for the project.

Selection Criteria	**Description**	**Weight (%)**
Strategic Alignment	Alignment with research objectives and national AI innovation goals	20%
Functional Fit	Ability to support ML-based risk prediction and Agentic AI decision support	20%
Technical Feasibility	Compatibility with selected technologies (React, Node.js, Firebase, Vertex AI)	15%
Cost Efficiency	Development and operational cost sustainability	15%

Usability & Accessibility	Ease of use for non-technical pet owners	10%
Scalability	Ability to expand features, users, and data volume	10%
Data Privacy & Security	Compliance with data protection and secure storage requirements	10%
Total		**100%**

b. Constraints and Limitations

The following constraints and limitations have been identified:

Technical Constraints

- Limited availability of large-scale, labeled pet symptom datasets.
- Dependence on survey-based data which may introduce bias.
- Internet connectivity required for real-time AI processing and clinic discovery.

Financial Constraints

- Limited research budget for cloud services and API usage.
- Reliance on free-tier or student-level technology resources.

Operational Constraints

- Availability of veterinary clinic data for integration.
- User adoption depends on digital literacy of pet owners.

Regulatory & Ethical Limitations

- The system provides risk prediction, not clinical diagnosis.
- Must ensure compliance with data privacy standards.
- AI recommendations must remain explainable and transparent.

5. Preferred Alternative

Development and implementation of a web-based, mobile-responsive Agentic AI-driven Decision Support System that uses machine learning and structured symptom data to provide early disease risk prediction and preventive guidance for companion animals.

a. Business Benefits

- Enables early disease risk detection, reducing severity and treatment costs.
- Improves preventive healthcare adoption through vaccination tracking and alerts.
- Enhances decision-making capability of pet owners through explainable AI guidance.
- Creates competitive differentiation compared to reminder-only or chatbot-based applications.
- Supports data-driven veterinary engagement, improving service accessibility.
- Strengthens potential for scalable commercialization within Sri Lanka's growing pet care market.

b. Cost

Cost Category	**Description**	**Type**
System Development	Frontend, backend, AI model development	One-time
Cloud Infrastructure	Hosting, database, API usage (Firebase, Vertex AI)	Recurring
API Integration	Google Maps API, notification services	Recurring
Testing & Quality Assurance	Model validation, usability testing	One-time
Maintenance & Updates	Bug fixes, model retraining, security updates	Recurring
Compliance & Data Protection	Privacy implementation and monitoring	Recurring

c. Assumptions

- Validated datasets can be used for initial ML training.
- Pet owners are willing to input symptom and vaccination data regularly.
- Internet connectivity is available for web-based access.

- Cloud infrastructure costs remain within projected limits.
- Veterinary clinics are willing to be listed within the system.

d. Analysis of Potential Impact on Business and Staff

Business Impact

- Establishes a technology-driven preventive pet healthcare solution.
- Creates potential revenue streams (subscription model, clinic partnerships).
- Enhances innovation positioning within the AI healthcare domain.

Staff Impact

- Requires technical expertise in AI model monitoring and system maintenance.
- Minimal operational staffing due to automation via Agentic AI.
- Training required for system administration and data monitoring.

e. Other Issues

- Regulatory considerations related to AI-based health advisory systems.
- Ethical responsibility to clearly state that the system does not replace veterinary diagnosis.
- Model accuracy limitations during early deployment phase.
- Dependence on third-party APIs (Google Maps, Firebase, AI services).

6. Risk Management Plan

a. Risk Assessment

Risk ID	Risk Description	Likelihood	Impact	Risk Level
R1	Insufficient or low-quality training data affecting ML accuracy	Medium	High	High
R2	Incorrect risk predictions leading to reduced user trust	Medium	High	High
R3	Data privacy or security breach	Low	High	High

R4	Integration issues with third-party APIs (Maps, Notifications)	Medium	Medium	Medium
R5	Low user adoption or resistance to AI-based recommendations	Medium	Medium	Medium
R6	Regulatory or ethical concerns regarding AI medical guidance	Low	High	Medium

b. Risk Response

Risk ID	**Mitigation Strategy**
R1	Use both primary (survey) and secondary validated datasets; apply data preprocessing and model validation techniques.
R2	Implement explainable AI outputs and clearly state that the system provides decision support, not medical diagnosis.
R3	Apply secure authentication, encryption, Firebase security rules, and compliance with data protection standards.
R4	Use well-documented APIs (Google Maps, Firebase) and conduct integration testing in early Agile sprints.
R5	Conduct usability testing and iterative refinement using Agile methodology.
R6	Include disclaimers and position the system strictly as a preventive decision support tool.

c. Benefit Realization

Benefits will be realized through:

- Improved early disease awareness among pet owners
- Reduction in delayed veterinary consultations
- Increased preventive care compliance (vaccination tracking)

- Enhanced user engagement with digital health monitoring
- Creation of a scalable AI-driven pet healthcare platform

Performance indicators:

- Model accuracy rate
- User adoption rate
- Vaccination reminder compliance rate
- User satisfaction feedback scores

7. Conclusion and Recommendations

The proposed Agentic AI-driven decision support system addresses a clear gap in companion animal preventive healthcare. Existing applications primarily focus on reminders or reactive consultation support, whereas this solution integrates machine learning-based early risk prediction, autonomous reasoning, and preventive care management within a mobile-accessible platform.

The project is technically feasible using modern AI frameworks and cloud-based architecture. From a business perspective, it offers strong potential for preventive healthcare value creation, cost efficiency for pet owners, and scalability within the digital veterinary support market.

Recommendations

- Proceed with Agile-based phased implementation.
- Prioritize development of a high-quality training dataset to ensure predictive accuracy.
- Ensure strict data privacy and security compliance from the initial development stage.
- Position the system clearly as a preventive decision support tool rather than a diagnostic substitute.
- Conduct pilot testing with a controlled group of pet owners before wider deployment.